Week 12 : Revision Lecture

F79MA Statistical Models A

Heriot-Watt University

Statistical parameter estimation concepts

Some definitions

A *statistic* is a random variable $T = T(\underline{X})$ which is a function of the data $\underline{X}$, but does not depend on θ.

An *estimator* is a statistic $T = T(\underline{X})$ such that the value of the realisation $t = T(\underline{x})$ is to be used as an estimate of θ.

The *bias* of an estimator $T = T(\underline{X})$ is defined by

$$B(T) = \mathbf{E}\,(T - \theta)\,.$$

T is *unbiased* for θ if $\mathbf{E}(T - \theta) = 0$ (or $\mathbf{E}(T) = \theta$).

T is *asymptotically unbiased* for θ if

$$B(T) \to 0 \qquad \text{as } n \to \infty\,.$$

Some definitions (continued)

The *mean square error* of T is defined to be

$$MSE(T) = \mathbf{E}\left[(T-\theta)^2\right] = \mathbf{Var}(T) + [B(T)]^2.$$

For estimators T_1 and T_2 we say that T_1 is *more efficient* than T_2 if

$$MSE(T_1) < MSE(T_2)$$

T is (mean-square) consistent for θ if

$$MSE(T) \to 0 \qquad \text{as } n \to \infty\,.$$

Method of moments in point estimation

Given a random sample $(X_1, \ldots, X_n)$, find parameter(s) $\widetilde{\theta}$ which make the (theoretical) population moments match the (observed) sample moments.

For models with a single parameter, the method-of-moments estimator (MME) is the value of $\widetilde{\theta}$ that solves

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i = \mu(\theta),$$

where $\mu(\cdot)$ is the population mean.

Interval estimation

A $100(1-\alpha)\%$ confidence interval for θ is a function of the data $\underline{X}$, i.e., $I(\underline{X}) = (\theta_L(\underline{X}), \theta_U(\underline{X}))$, such that

$$\mathbf{P}\left(\theta \in I(\underline{X})\right) = 1 - \alpha,$$

where α is suitably small (often 0.05).

Note that $\underline{X}$ is a random vector, so $I(\underline{X})$ is a *random interval*.

We say that $100(1-\alpha)\%$ of the random samples from the population of $I(\underline{X})$ contain the true value of θ, or we are $100(1-\alpha)\%$ confident that the random interval $I(\underline{X})$ contains θ.

Pivotal quantities

A *pivotal quantity* or *pivot* is a function of the data $\underline{X}$ and the parameter θ such that

- its distribution is known, and
- its distribution does not depend on θ.

To find a $100(1-\alpha)\%$ confidence interval for θ, we write

$$\mathbf{P}\left(z_1 < Z(\underline{X}, \theta) < z_2\right) = 1 - \alpha,$$

for a pivotal quantity $Z(\underline{X}, \theta)$.

When $Z(\underline{X}, \theta)$ is a monotonic function of θ, we rearrange the inequality to

$$\mathbf{P}\left(\theta_L(\underline{X}) < \theta < \theta_U(\underline{X})\right) = 1 - \alpha.$$

The likelihood function, estimation, and the likelihood principle

The likelihood function

The likelihood function of θ (given the observed data $\underline{x}$) is

$$L(\theta) = L(\theta; \underline{x}) = f_{\underline{X}}(\underline{x}; \theta),$$

where $f_{\underline{X}}$ is the joint mass or density function of $\underline{X}$.

If $\underline{x} = (x_1, \ldots, x_n)$, where the x_i's are i.i.d, then

$$L(\theta; \underline{x}) = \prod_{i=1}^{n} f_X(x_i; \theta),$$

where f_X denotes the mass or density function of X.

The maximum likelihood estimator

The *maximum likelihood estimator* (MLE) for θ, denoted by $\hat{\theta}$, is the value of θ that maximises $L(\theta; \underline{X})$.

When we observe an outcome $\underline{x}$ of $\underline{X}$, we have the maximum likelihood estimate $\widehat{\theta}(\underline{x})$, which is an outcome of the estimator $\widehat{\theta}(\underline{X})$.

To derive the MLE, it is usually easier in practice to maximise $l(\theta) = \log L(\theta)$ rather than $L(\theta)$.

Invariance property of MLEs: If the function g is bijective (one-to-one correspondence), then the MLE of the parametrisation $\phi = g(\theta)$ is $\hat{\phi} = g(\widehat{\theta})$.

The likelihood principle

If $\underline{x}$ and $\underline{y}$ are two possible experimental outcomes such that $L(\theta; \underline{x})$ is proportional to $L(\theta; \underline{y})$, i.e.

$$L(\theta; \underline{x}) = C(\underline{x}, \underline{y}) L(\theta; \underline{y}),$$

then the conclusions about θ drawn from $\underline{x}$ and $\underline{y}$ should be identical.

Sufficiency

For an experiment with outcome $\underline{X}$, a statistic $T = T(\underline{X})$ is *sufficient* for θ if the conditional distribution of $\underline{X}$ given the value of T does not depend on θ, i.e.

$$f_{\underline{X}|T}(\underline{x}|t(\underline{x}) = t) = h(\underline{x}) \qquad \forall\theta\,.$$

The Factorisation Theorem: $T = T(\underline{X})$ is sufficient for θ if and only if

$$f_{\underline{X}}(\underline{x};\theta) = h_1(\underline{x})h_2(t(\underline{x}),\theta),$$

for some functions h_1 and h_2.

The factorisation theorem is useful in the application of the likelihood principle. We draw the same inference about θ if, for two experimental outcomes $\underline{x}$ and $\underline{y}$, we have $t(\underline{x}) = t(\underline{y})$.

The Cramer-Rao lower bound and the variance of unbiased estimators

The score function

Let $\underline{X}$ be a random sample from a distribution with joint density $f_{\underline{X}}(\underline{x};\theta)$. We define the score function as

$$U(\theta, \underline{X}) = \frac{\partial}{\partial \theta} \log f_{\underline{X}}(\underline{x};\theta).$$

Some important properties:

- $\mathbf{E}(U) = 0$
- $\mathbf{Var}(U) = \mathbf{E}(U^2)$

The Information $I(\theta)$

The amount of information about θ is defined as

$$I(\theta) \triangleq \mathbf{E}\left(\left[\frac{\partial}{\partial\theta}\log f_{\underline{X}}(\underline{x};\theta)\right]^2\right) = \mathbf{E}(U^2).$$

Under some regularity conditions, we can use the equivalent expression

$$I(\theta) = -\mathbf{E}\left(\frac{\partial^2}{\partial\theta^2}\log f_{\underline{X}}(\underline{x};\theta)\right) = -\mathbf{E}\left(\frac{\partial}{\partial\theta}U\right).$$

The Cramer-Rao Lower Bound

Let $T(\underline{X})$ be an unbiased estimator of θ.

Under some regularity conditions on $f_{\underline{X}}(\underline{x};\theta)$ and $T(\underline{X})$, we have the Cramer-Rao lower bound (CRLB) as

$$\textbf{Var}\left(T(\underline{X})\right) \geq \frac{1}{I(\theta)} = \frac{1}{\textbf{E}\left(\left[\frac{\partial}{\partial\theta}\log f_{\underline{X}}(\underline{x};\theta)\right]^2\right)}.$$

Under some additional regularity conditions, the bound can be expressed as

$$\textbf{Var}\left(T(\underline{X})\right) \geq \frac{-1}{\textbf{E}\left(\frac{\partial^2}{\partial\theta^2}\log f_{\underline{X}}(\underline{x};\theta)\right)}.$$

The Cramer-Rao Lower Bound (continued)

Under suitable regularity conditions, if the elements of the sample $\underline{X} = (X_1, \ldots, X_n)$ are i.i.d given θ, then

$$\mathbf{E}\left(\left[\frac{\partial}{\partial\theta}\log f_{\underline{X}}(\underline{x};\theta)\right]^2\right) = -\sum_{i=1}^{n}\mathbf{E}\left(\frac{\partial^2}{\partial\theta^2}\log f_{X_i}(x_i;\theta)\right)$$

$$= -n\mathbf{E}\left(\frac{\partial^2}{\partial\theta^2}\log f_X(x;\theta)\right)$$

Minimum variance unbiased estimators

Minimum variance unbiased estimators (MVUE) are unbiased estimators that have the smallest variance for a specific problem.

If an unbiased estimator attains the CRLB, then it must be the MVUE.

Let $\underline{X}$ be a random sample from a distribution satisfying the regularity conditions of the CRLB. Let $T(\underline{X})$ be a smooth unbiased estimator of θ.

CRLB Corollary: $T(\underline{X})$ attains the CRLB if and only if $U(\theta, \underline{X})$ can be expressed in the form

$$U(\theta, \underline{X}) = a(\theta)\,[T(\underline{X}) - \theta]$$

for some function $a(\theta)$.

Consequences of the CRLB Corollary

If $U(\theta, \underline{X})$ can be written in the form

$$U(\theta, \underline{X}) = a(\theta)\,[T(\underline{X}) - \theta]\,,$$

where $T(\underline{X})$ is any statistic, then

(i) T is unbiased for θ.

(ii) T attains the CRLB.

(iii) T is the MLE of θ.

(iv) $a(\theta) = I(\theta)$, the information, so that $\mathbf{Var}\,(T) = 1/a(\theta)$.

The MLE $\widehat{\theta} = T(\underline{X})$ is the MVUE if and only if $U(\theta, \underline{X})$ can be expressed in the form

$$U(\theta, \underline{X}) = a(\theta)\,[T(\underline{X}) - \theta]\,.$$

Consequences of the CRLB Corollary (continued)

Uniqueness: There can only be one way of writing the score function in the form

$$U(\theta, \underline{X}) = a(\theta)\,[T(\underline{X}) - \theta]\ ,$$

so $T = T(\underline{X})$ must be the *unique* MVUE if such a representation of the score function exists.

Parameter transformations: Suppose $\phi = \phi(\theta)$ is a function of θ. Then the existence of a MVUE for θ that attains the CRLB *does not imply* the existence of a similar MVUE for ϕ, unless $\phi(\theta)$ is a linear function.

CRLB for biased estimators

If $T(\underline{X})$ is a biased estimator for θ, then the CRLB is

$$\begin{aligned}\mathbf{Var}\left(T(\underline{X})\right) &\geq \frac{\left(\frac{\partial}{\partial\theta}\mathbf{E}\left[T(\underline{X})\right]\right)^2}{n\mathbf{E}\left(\left[\frac{\partial}{\partial\theta}\log f_X(x;\theta)\right]^2\right)} \\ &= \frac{\left[1+\frac{\partial}{\partial\theta}B(T)\right]^2}{n\mathbf{E}\left(\left[\frac{\partial}{\partial\theta}\log f_X(x;\theta)\right]^2\right)}\end{aligned}$$

where $B(T)$ is the bias of T, since $\mathbf{E}[T(\underline{X})] = \theta + B(T)$.

Asymptotic properties of maximum likelihood estimators and the deviance function

Asymptotic normality of the MLE

Let $\underline{X}_n$ be an i.i.d. random sample of size n from a distribution with density function $f_X(x;\theta)$ verifying appropriate regularity conditions, and let $\widehat{\theta}(\underline{X}_n)$ be the MLE computed from $\underline{X}_n$.

Then,

$$\frac{\widehat{\theta}(\underline{X}_n) - \theta}{\sqrt{I(\theta)^{-1}}} \to \mathrm{N}(0,1)\,, \quad \text{as } n \to \infty.$$

We say that, **asymptotically**, MLEs:

- are unbiased,
- are normally distributed, and
- attain the CRLB.

Asymptotic normality of the MLE (continued)

We often apply

$$\frac{\widehat{\theta}(\underline{X}_n) - \theta}{\sqrt{I(\theta)^{-1}}} \sim \mathrm{N}(0, 1),$$

as an approximation when the sample size n is large.

We use this approximation to obtain a pivotal quantity for θ to construct an approximate 95% confidence interval for θ from the MLE:

$$\widehat{\theta} \pm 1.96\sqrt{I(\widehat{\theta})^{-1}},$$

where $I(\theta)$ is approximated by $I(\widehat{\theta})$ in the denominator of the pivot.

Asymptotic properties of the deviance function

The deviance function for any θ is given by

$$D(\theta) = 2\left[l(\widehat{\theta}(\underline{X}_n)) - l(\theta)\right],$$

where $\widehat{\theta}(\underline{X}_n)$ is the MLE of θ computed from $\underline{X}_n$.

As $n \to \infty$, the deviance $D(\theta)$ converges to a χ^2_1 random variable.

Corollary: Suppose the value $\chi^2_1(\alpha)$ is such that

$$\mathbf{P}(\chi^2_1 > \chi^2_1(\alpha)) = \alpha.$$

Then an approximate large-sample $100(1-\alpha)\%$ confidence interval for θ is

$$\left\{\theta : D(\theta) < \chi^2_1(\alpha)\right\}.$$

Hypothesis testing

Definition of hypotheses

A *hypothesis* is a statement about a parameter θ with regard to a subset of the parameter space Ω.

Hypothesis tests typically involve two competing hypotheses: the *null hypothesis*

$$H_0 : \theta \in \omega,$$

and the *alternative hypothesis*

$$H_1 : \theta \in \omega' = \Omega \setminus \omega.$$

The value of θ is unknown, so we do not know which hypothesis is true.

Definition of hypothesis tests

To conduct the hypothesis test, we observe some data $\underline{x}$ related to θ by a statistical model with probability function $f_{\underline{X}}(\underline{x}; \theta)$.

A hypothesis test is a rule for partitioning the sample space (for the data $\underline{x}$) into two regions, as follows:

- A critical or rejection region, denoted R, such that if $\underline{x} \in R$ then we **reject** H_0,
- And its complement, R', such that if $\underline{x} \in R'$, then we **fail to reject** H_0.

Note that we say "fail to reject H_0" instead of "accept H_0".

Error probabilities of a hypothesis test

Type I error: A Type I error occurs when we incorrectly reject H_0, although it is actually true.

The probability of a Type I error is

$$\alpha(\theta) = \mathbf{P}(\underline{X} \in R), \theta \in \omega.$$

Type II error: A Type II error occurs when we fail to reject H_0, although it is false. The probability of a Type II error is

$$\beta(\theta) = \mathbf{P}(\underline{X} \in R'), \theta \in \omega'.$$

$1 - \beta(\theta)$ is known as the *power* function of the test.

A good hypothesis test

A good hypothesis test specifies R and R' in a way that minimises both α and β.

The standard procedure for designing a test is to first fix the value of α (usually a small value like 5%).

The chosen value of α is known as the *size* or *level* of the test.

Having fixed the value of α, we then try to find the test which minimises β, or equivalently, maximises the power 1–β subject to attaining the level α.

The Neyman-Pearson Lemma

Suppose we have two simple hypotheses

$$H_0 : \theta = \theta_0 \ v. \ H_1 : \theta = \theta_1.$$

The Neyman-Pearson (NP) Lemma tells us that the most powerful (MP) test of size α is such that

$$R = \left\{ \underline{x} \;\middle|\; \frac{f_{\underline{X}}(\underline{x}; \theta_0)}{f_{\underline{X}}(\underline{x}; \theta_1)} \leq k \right\},$$

where k depends on the size of the test.

The smaller α, the smaller k must be.

Step-by-step procedure

Applying the NP Lemma to construct optimal tests for two simple competing hypotheses involves the following steps

1. First we construct the ratio $\frac{f_{\underline{X}}(\underline{x};\theta_0)}{f_{\underline{X}}(\underline{x};\theta_1)}$ and the associated critical region R.

2. We re-express R in the form $\{\underline{x} : s(\underline{x}) \leq k^* \text{ (or } \geq k^*)\}$ where $s(\underline{x})$ is a function of $\underline{x}$ whose sampling distribution is known under H_0 and H_1.

3. We use the sampling distribution of $s(\underline{X})$ under H_0 to identify the required value of k^* to give the desired error rate α.

Uniformly most powerful tests for composite hypotheses

When a single test verifies the NP Lemma for all $\theta \in \omega'$, we refer to this test as the Uniform Most Powerful Test (UMP).

For one-sided tests of the form

$$H_0 : \theta = \theta_0 \ v. \ H_1 : \theta > \theta_0. \quad \text{or} \quad H_0 : \theta = \theta_0 \ v. \ H_1 : \theta < \theta_0,$$

a UMP exists if the population distribution has the *Monotone Likelihood Ratio Property* (MLRP).

A family of pdf's or pmf's has the MLRP if there exists a statistic $g(\underline{x})$ such that for every $\theta_1 > \theta_0$, $\frac{f_{\underline{X}}(\underline{x};\theta_0)}{f_{\underline{X}}(\underline{x};\theta_1)}$ is a strictly monotonic increasing function of $g(\underline{x})$.

Exponential family of distributions

A random variable has a distribution from the exponential family if we can write the probability density or mass function in the form

$$f(x;\theta) = c(\theta)h(x)\exp\{q(\theta)t(x)\}$$

for some functions c, h, q and t.

Such random variables will have the MLRP as long as $q(\theta)$ is strictly monotonic.

Probability-values or p-values

The p-value is the minimum level of test at which H_0 would be rejected given the observed data, or equivalently, as the probability of observing data 'as extreme as' the given data when H_0 is true.

There is a natural connection between p-values and testing at fixed level α - if the p-value is less than α, then reject at level α.

The p-value "quantifies the evidence" against H_0 - the smaller the p-value the stronger the evidence against H_0.

Generalised likelihood ratio tests

UMP tests do not exist for two-sided alternative hypotheses

$$H_0 : \theta = \theta_0 \ v. \ H_1 : \theta \neq \theta_0.$$

The rejection region for the generalised likelihood ratio test (GLRT) is

$$R = \left\{ \underline{x} \;\middle|\; \frac{f_{\underline{X}}(\underline{x}; \hat{\theta}_0)}{f_{\underline{X}}(\underline{x}; \hat{\theta})} \leq k \right\},$$

where $\hat{\theta}_0$ and $\hat{\theta}_1$ are the values of θ that maximise the likelihood over ω and ω' respectively.

The value of k can be determined for a requisite size α based on the sampling distribution of $\frac{f_{\underline{X}}(\underline{x};\hat{\theta}_0)}{f_{\underline{X}}(\underline{x};\hat{\theta})}$.

Asymptotic result for the GLRT

For a sufficiently large sample size n, we have (approximately):

$$-2 \log \lambda(\underline{X}) \sim \chi^2_d,$$

where

$$\lambda(\underline{x}) = \frac{f_{\underline{X}}(\underline{x}; \hat{\theta}_0)}{f_{\underline{X}}(\underline{x}; \hat{\theta})},$$

and d is the difference in dimensionality between the parameter space Ω and the dimensionality of the set ω.

Bayesian statistical inference

The concept of Bayesian inference

Suppose that we carry out an experiment where the outcome $\underline{x}$ has probability function $f(\underline{x};\theta)$, where θ is an unknown parameter of interest. Suppose our prior belief about θ is described by the probability density/ mass function $\pi(\theta)$.

Then, on observing $\underline{x}$, the posterior distribution of θ is given using Bayes' theorem as

$$\pi(\theta|\underline{x}) = \frac{\pi(\theta)L(\theta;\underline{x})}{\int \pi(\theta')L(\theta';\underline{x})d\theta'},$$

where $L(\theta;\underline{x}) = f(\underline{x};\theta) = \pi(\underline{x}|\theta)$ is the likelihood.

If $\pi(\theta)$ is a discrete distribution, then the integral on the denominator just becomes a sum.

Step-by-step

- **Step 1:** Specify a prior distribution for θ representing belief about θ before any observations are taken.
- **Step 2:** Carry out an experiment and collect a sample of n observations $\underline{x} = (x_1, \ldots, x_n)$. Derive the likelihood function $L(\theta; \underline{x})$ of observing the sample $\underline{x}$.
- **Step 3:** Derive the posterior distribution of θ given $\underline{x}$. One can often determine the integral/ sum on the denominator by recognising that the algebraic form of the posterior density/ mass resembles those of some standard distribution.

Conjugate priors

A conjugate prior is one that shares the same distribution as the posterior after combining with the likelihood function.

Some examples seen in class:

- The Beta distribution is the conjugate prior for p in Bin(n, p).
- The Gamma distribution is the conjugate prior for λ in Exp(λ).
- The Gamma distribution is the conjugate prior for λ in Poi(λ).
- The Normal distribution is the conjugate prior for μ in N(μ, σ^2) where σ^2 is known.

Summarising posterior distributions

Apart from graphical representations of the posterior density, we often also report the posterior mean and variance. We can also report a $100(1-\alpha)\%$ credible interval for θ, i.e. (θ_L, θ_U) so that

$$\mathbf{P}\left(\theta \in (\theta_L, \theta_U) \mid \underline{x}\right) = \int_{\theta_L}^{\theta_U} \pi(\theta|\underline{x}) d\theta = 1 - \alpha.$$

Equi-tailed credible interval: Exclude an area of $\frac{\alpha}{2}$ from each tail of the posterior distribution.

Highest posterior density credible region: Identify a value $\gamma_\alpha > 0$ such that the region $\mathcal{C}_\alpha = \{\theta : \pi(\theta|\underline{x}) > \gamma_\alpha\}$ satisfies

$$\int_{\mathcal{C}_\alpha} \pi(\theta|\underline{x}) d\theta = 1 - \alpha.$$

Choice of priors

Informative priors

- Informative priors are specified in order to reflect strong prior information or beliefs about a parameter θ.
- We generally want $\mathbf{Var}(\theta)$ to be small.

Vague priors

- Vague priors 'support' a wide range of values of the parameter θ and does not strongly influence our inferences.
- We generally want $\mathbf{Var}(\theta)$ to be large.

Jeffreys' prior

The Jeffreys' prior for an unknown parameter θ is given by

$$\pi_J(\theta) \propto I(\theta)^{\frac{1}{2}},$$

where $I(\theta)$ denotes the Fisher information.

Jeffreys' priors are invariant to changes of parametrisation (1-1 transformations).

Jeffreys' priors are often *improper* and are hence only specified up to a proportionality constant.

Posterior predictive distributions

Suppose that we have updated our posterior density/ mass function $\pi(\theta; \underline{x})$ from our prior $\pi(\theta)$ and likelihood of observed data $L(\theta; \underline{x})$.

The posterior predictive density/ mass function for a future random variable Z given $\underline{x}$ is

$$\pi(z|\underline{x}) = \int \pi(z|\theta) \cdot \pi(\theta|\underline{x}) d\theta,$$

where $\pi(z|\theta)$ is the likelihood of the future observation z.

The resulting predictive distribution is often a mixture distribution, e.g. the beta-binomial, gamma-Poisson distributions etc.

Bayes principle in decision theory

Suppose that we have a set of possible actions $\{a_1, a_2, \ldots, a_k\}$ and a set of possible state of natures $\{\theta_1, \theta_2, \ldots, \theta_m\}$. Denote by $L(\theta_i, a_j)$ the loss incurred by choosing action a_j when state θ_i is true, where $i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, k$.

Suppose that we describe our uncertainty about θ using $\pi(\theta_i|\underline{x})$, for $i = 1, 2, \ldots, m$.

The Bayes principle is to select the action a_j which minimises the expected posterior loss, which is defined as

$$B(a_j) = E_\theta[L(\theta, a_j)] = \sum_i L(\theta_i, a_j)\pi(\theta_i|\underline{x}).$$

In the case where θ takes continuous values, we replace the sum by an integral.

Bayesian hypothesis testing (simple hypotheses)

Consider the simple hypotheses $H_0 : \theta = \theta_0$ *v.* $H_1 : \theta = \theta_1$. Let a_1 be the action to not reject H_0, and a_2 be the action to reject H_0. Suppose that the loss table is

	a_1	a_2
θ_0	L_{00}	L_{01}
θ_1	L_{10}	L_{11}

According to the Bayes principle, we choose a_1 if

$$\frac{f(\underline{x};\theta_0)}{f(\underline{x};\theta_1)} > \frac{\pi(\theta_1)L_{10}}{\pi(\theta_0)L_{01}},$$

which resembles the likelihood ratio test in the classical setting. Here, $\pi(\theta_0)$ and $\pi(\theta_1)$ are the prior probabilities for θ_0 and θ_1.

Bayesian point estimation

Suppose that we have observed some data $\underline{x}$ and derived a posterior distribution $\pi(\theta|\underline{x})$.

We need to choose what value $\hat{\theta}$ to report as an estimate of $\theta|\underline{x}$.

According to Bayes principle, when the loss is described as a quadratic loss/ mean square error loss, i.e.

$$L(\theta, a) = (a - \theta)^2,$$

then the Bayes estimate is the posterior expectation, i.e.

$$\hat{\theta} = \mathbf{E}(\theta|\underline{x})$$

Credibility theory

Credibility estimate

The credibility estimate for a risk can be expressed as

$$Z\bar{X} + (1 - Z)\mu,$$

where $0 \leq Z \leq 1$ is the credibility factor.

In particular,

- $\bar{X}$ can be thought of as the mean of data directly related to the risk, and
- μ can be thought of as the mean of data from a larger group of similar, but not necessarily identical, risks (known as "collateral data")

Bayesian approach to credibility - Poisson/gamma model

Let X denote the number of claims in the coming year from a certain risk. and $X \sim \text{Poi}(\lambda)$. Suppose that we choose the prior $\lambda \sim \text{Gamma}(\alpha, \beta)$. Suppose further that $\underline{x} = (x_1, \cdots, x_n)$ is collected.

Then the Bayes estimate under mean square error loss (i.e. the posterior expectation) is

$$\mathbf{E}(\lambda|\underline{x}) = Z\bar{x} + (1 - Z)\mu,$$

where

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \ , \quad Z = \frac{n}{\beta + n} \ , \quad \text{and} \quad \mu = \frac{\alpha}{\beta}.$$

Bayesian approach to credibility - Normal/normal model

Let X denote the monetary amount of the aggregate claim in the coming year from a certain risk, and $X \sim \mathrm{N}(\theta, \sigma_1^2)$. Suppose that we choose the prior $\theta \sim \mathrm{N}(\mu, \sigma_2^2)$. Suppose further that $\underline{x} = (x_1, \cdots, x_n)$ is collected.

Then the Bayes estimate under mean square error loss (i.e. the posterior expectation) is

$$\mathbf{E}(\lambda|\underline{x}) = Z\bar{x} + (1 - Z)\mu,$$

where

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \quad \text{and} \quad Z = \frac{n}{n + \frac{\sigma_1^2}{\sigma_2^2}} = \frac{n}{n + \frac{\text{within risk variance}}{\text{between risk variance}}}.$$

Bühlmann credibility formula

The credibility estimate for risk i takes the form

$$Z\bar{X}_i + (1 - Z)\mu,$$

where $\bar{X}_i = \frac{1}{n}\sum_{j=1}^{n} X_{ij}$.

The credibility factor is calculated as

$$Z = \frac{n}{n + \frac{\mathbf{E}[s^2(\theta_i)]}{\mathbf{Var}[m(\theta_i)]}}.$$

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The credibility factor is calculated as

$$Z = \frac{n}{n + \frac{\mathbf{E}[s^2(\theta_i)]}{\mathbf{Var}[m(\theta_i)]}}.$$

Empirical Bayes Credibility Theory Model 1

When a parametric model is not available, we estimate the structural parameters using suitable estimators.

$$\widehat{\mathbf{E}[m(\theta_i)]} = \bar{X} = \frac{1}{N}\sum_{i=1}^{N}\bar{X}_i = \frac{1}{nN}\sum_{i=1}^{N}\sum_{j=1}^{n}X_{ij}$$

$$\widehat{\mathbf{E}[s^2(\theta_i)]} = \frac{1}{N}\sum_{i=1}^{N}S_i^2 = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1}{n-1}\sum_{j=1}^{n}\left(X_{ij}-\bar{X}_i\right)^2\right]$$

$$\widehat{\mathbf{Var}[m(\theta_i)]} = \frac{1}{N-1}\sum_{i=1}^{N}\left(\bar{X}_i-\bar{X}\right)^2 - \frac{1}{nN}\sum_{i=1}^{N}\left[\frac{1}{n-1}\sum_{j=1}^{n}\left(X_{ij}-\bar{X}_i\right)^2\right]$$

Likelihood and Maximum Likelihood Estimation in More Complex Situations

Examples of complex situations and some strategies

Likelihood functions of transformed parameters

- It is very important to take into account the *range* of a transformed parameter when considering maximum likelihood estimation.
- The estimate derived from unconstrained maximisation of the likelihood may lie outside the permissible range.
- The likelihood function should be zero for parameter values lying outside of the permissible range, e.g., for the $\mathrm{U}(0, \theta)$ distribution, $L(\theta; \underline{x}) = 0$ for $\theta < x_{(n)}$.

Examples of complex situations and some strategies (continued)

Solving for MLEs using non-analytic methods

- In certain situations, we might derive an equation involving the parameter that cannot be solved analytically.
- We might need to resort to numerical root-finding methods, e.g. Newton-Raphson method, Bisection method, secant method, etc.
- Using R to implement these iterative procedures would be helpful.

Examples of complex situations and some strategies (continued)

Introducing nuisance parameters in our model

- Oftentimes we may need to create additional parameters whose inference is not of interest to us.
- A useful observation is that these (nuisance) parameters can often be expressed as a 1-1 function of our parameter of interest.
- In such cases, the invariance property of MLEs would come in handy.

Examples of complex situations and some strategies (continued)

Dealing with censored or truncated data

- When fitting statistical models to lifetimes, survival times, etc., we often find that data measurement ceases before all the times have been observed.
- It is important to consider the likelihoods of each individual component in the experiment, including those that have not been realised at the end of the experiment.

Examples of complex situations and some strategies (continued)

Observations that are not identically distributed

- In some cases, we may have observations that come from varying density/ mass functions.
- As long as the observations are independent, we can derive a likelihood function for our parameter as a product of individual density/ mass functions and maximise it.
- One example seen in class is the simple birth process (Yule process), where the inter-birth time intervals follow an exponential distribution with a rate that increases with time.